



VANSHITA ARYA

+91 9694222811

vanshita.arya@outlook.com

[Vanshita_Linkedin](#)

Core Competencies

Data Science & Machine Learning

Cybersecurity & Fraud Analytics

Statistical Analysis & Forecasting

NLP & Text Analytics

Data Visualization & Reporting

Model Deployment & Automation

Financial & Risk Analytics

Version Control

Technical Skills

Programming Languages: Python (Intermediate) | SQL (Basic)

Machine Learning & AI: Scikit-learn | SciPy | NLP (VADER, TextBlob, and RegEx) | ML |

Libraries & Modules: Pandas | NumPy | Matplotlib | Seaborn

Data Visualization: Excel (Intermediate) | Power BI (Basic)

Web Development & Frameworks: FastAPI (Backened) | Streamlit (Frontened)

Tools & Frameworks: Git | CBS Finnacle

Education

- 2023 - Present
M.Sc. in Data Science
Amity University, Noida
CGPA: 9.13 (till 3rd SEM)
- 2017 - 2020
B.Sc. in Maths, Statistics, Physics
Banasthali Vidyapith
CGPA: 7.6

Social Links

[GitHub](#)
[kaggle](#)
[Newsletter](#)

Profile Summary

- Data Science Professional with Hands-On Experience:** Strong background in Exploratory Data Analysis, predictive analytics, and fraud detection, acquired through industry internships and academic projects.
- Expert in Cyber Fraud Analysis & Financial Crime Investigation:** Assisted law enforcement and financial institutions in identifying fraud patterns and money laundering risks using advanced data techniques.
- Proven Ability to Develop Impactful Data Solutions:** Designed and implemented data-driven systems such as fraud response automation and stock market analytics.
- Passionate About Data-Driven Decision Making:** Skilled in deriving meaningful insights from complex datasets, enabling businesses and agencies to make informed choices.
- Collaborative & Research-Oriented Mindset:** Adept at working in interdisciplinary teams and continuously learning new methodologies in AI and analytics.

Work Experience

Dec'24 – Feb'25 | **Cyber Crime Data Analyst Intern** | **I4C, Ministry of Home Affairs**

- Cyber Fraud Analysis & Investigation:** Analyzed **NCRP portal data** to detect fraud patterns in **UPI, e-wallets, cryptocurrency, and CBDC**. Identified **money laundering hotspots** and assisted in **cross-verifying cyber complaints** for risk mitigation.
- Data Extraction & Visualization:** Aggregated **financial data** and built **solution** to track **ATM withdrawal hotspots, fraud trends, and bank performance**.
- Geospatial & NLP Processing:** Used **geospatial mapping** to analyze fraud locations and applied **NLP & RegEx** to extract regional insights from complaint data.
- Collaboration & Compliance:** Partnered with **Telecom Service Providers (TSPs)** to **block fraudulent mobile numbers** and worked with **banks & law enforcement agencies** to strengthen cybersecurity frameworks.

Mar'24 | **Data Science Intern** | **British Airways**

- Extracted & Preprocessed 10K+ customer reviews** using **BeautifulSoup (web scraping), tokenization, lemmatization, and stopword removal**.
- Applied Sentiment Analysis** using **VADER & TextBlob** to detect customer sentiment trends.
- Used Word2Vec & Topic Modeling** to analyze customer feedback and generate insights.
- Developed a Random Forest Model** to predict customer holiday purchases, improving classification accuracy.
- Built Data Visualizations** using **Seaborn & Matplotlib** to present findings for better business decisions.

Projects

Fraud Intelligence & Response Automation (FIRA) | Python, Excel, Power BI:

- Developed an automated **fraud detection system** and a **real-time complaint tracking System**, improving fraud detection efficiency by **40%** and reducing response time for financial fraud cases.

National Stock Exchange Data Analysis | Python, Streamlit, Kaggle API

- Designed a **self-service BI platform** using **Streamlit** for analyzing stock performances on **NSE India**, providing real-time data insights.
- Published a Kaggle dataset with weekly updates, achieving a high usability score of 8.82

Certificate and Courses

- Azure Fundamentals** (Microsoft)
- Introduction to Git and GitHub** (Google)
- Essentials of Data Science with R – Probability and Statistical Inference** (IIT Kanpur)
- Everyday Excel** (University of Colorado Boulder)

